

Airline Management System

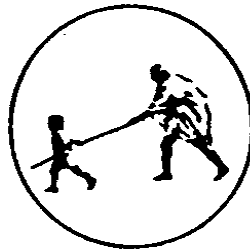
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Under the Guidance

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Mahatma Gandhi Mission's College of Engineering, Nanded (M.S.)

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Airline Management System

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BACHELOR OF TECHNOLOGY
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of

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING
NANDED (M.S.)

Academic Year 2025-26

Certificate



This is to certify that the project entitled

“Airline Management System”

*being submitted by **Sumit Piraji Balkhande** to the Dr. Babasaheb Ambedkar Technological University, Lonere, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by him/her under my supervision and guidance. The matter contained in this report has not been submitted to any other university or institute for the award of any degree.*

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ABSTRACT

This project outlines the design and implementation of an Airline Management System using a relational **Database Management System (DBMS)**. The primary objective is to develop a robust, efficient, and centralized system to manage the complex and interconnected data involved in modern airline operations. The system is designed to handle core functions, including flight scheduling, passenger reservation, ticketing, and staff management. Key entities modelled include **Flights** (with attributes such as departure/arrival airports, times, and aircraft ID), **Passengers** (personal details and contact information), **Bookings** (linking passengers to specific flights and seats), and **Employees** (pilots, crew, and ground staff with assigned duties).

The system will utilize an Entity-Relationship (ER) **model** for conceptual design, which will then be mapped to a relational schema, implemented using SQL, and normalized to ensure data integrity, minimize redundancy, and optimize transaction performance.

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Chapter1

Introduction

The Airline Management System is a database-driven application designed to streamline and automate the operations of an airline. Modern airline services involve the management of flights, passengers, reservations, staff, schedules, and payment records—tasks that become increasingly complex as the volume of data grows. A centralized database system helps maintain accurate, consistent, and reliable information, ensuring that airline operations run smoothly. By using database concept

1.1 HISTORY

The concept of airline management systems dates to the early days of commercial aviation, when airlines manually recorded passenger information, flight schedules, and ticket bookings using paper-based registers. As the aviation industry expanded in the mid-20th century, manual methods became inefficient and error prone.

1.2 METHODOLOGY

The methodology used for developing the Airline Management System follows a structured and systematic approach to ensure efficient database design and smooth system functionality. The process begins with requirement analysis, where user needs such as flight management, passenger records, booking details, and staff information are identified.

1.3 SYSTEM REQUIREMENTS

Hardware Requirements :

The Airline Management System requires a basic yet efficient hardware setup capable of handling database operations and user interactions. A computer system with at least an Intel i3 or equivalent processor.

Software Requirements :

The system is developed using database management tools and programming environments that support reliable data storage and manipulation. Required software includes an operating system such as Windows, Linux, or macOS, along with a DBMS like MySQL, PostgreSQL, Oracle, or SQL Server for managing airline data

CHAPTER 2

System design

The System Design of the Airline Management System focuses on creating a structured and efficient database architecture that supports all major airline operations such as flight scheduling, passenger management, ticket booking, and staff handling. The design process begins by identifying key entities like Passengers, Flights, Bookings, Payments, and Staff, followed by establishing their relationships using an Entity–Relationship (ER) Diagram.

2.1 ER DIAGRAM

Entities and Attributes

1. Passenger

- **Attributes:**

Passenger ID (PK), Name, Email, Phone, Address, Passport Number, Age

2. Flight

- **Attributes:**

Flight ID (PK), Flight Number, Source, Destination, Departure Time, Arrival Time, Seats Available, Status

3. Booking

- **Attributes:**

Booking ID (PK), Passenger ID (FK), Flight ID (FK), Booking Date, Seat Number, Class, Booking Status

4. Payment

- **Attributes:**

Payment ID (PK), Booking ID (FK), Amount, Payment Method, Payment Date, Payment Status

2.2 DATA FLOW

DFD – Level 0 (Context Diagram)

The Level 0 DFD represents the Airline Management System as a single high-level process interacting with external entities. The main external users include **Passenger**, **Admin/Staff**, and the **Payment Gateway**. Passengers interact with the system to search flights, make bookings, and view ticket details. Admins manage flight schedules, passenger data, and staff information. The Payment Gateway handles online payments for bookings. The Airline Management System processes all incoming data, stores it in the database, and returns outputs such as booking confirmation, flight details, receipts, and updated records.

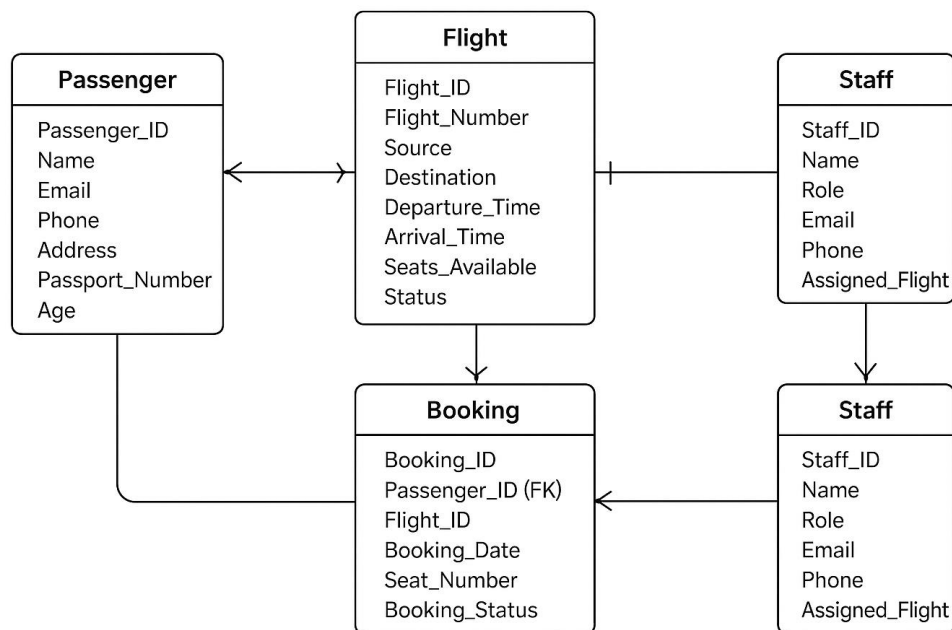
DFD – Level 1 (Process Breakdown)

The Airline Management System is divided into the following major processes:

1. Manage Passengers

- Input: Passenger details

ER Diagram - Airline Management System



2.2 ER Diagram

2. Manage Flights

- Input: Flight schedule, route details
- Output: Updated flight information

3. Booking Process

- Input: Passenger ID, Flight ID, seat selection
- Output: Booking confirmation

4. Payment Processing

- Input: Transaction details
- Output: Payment receipt / status
- External Entity: *Payment Gateway*

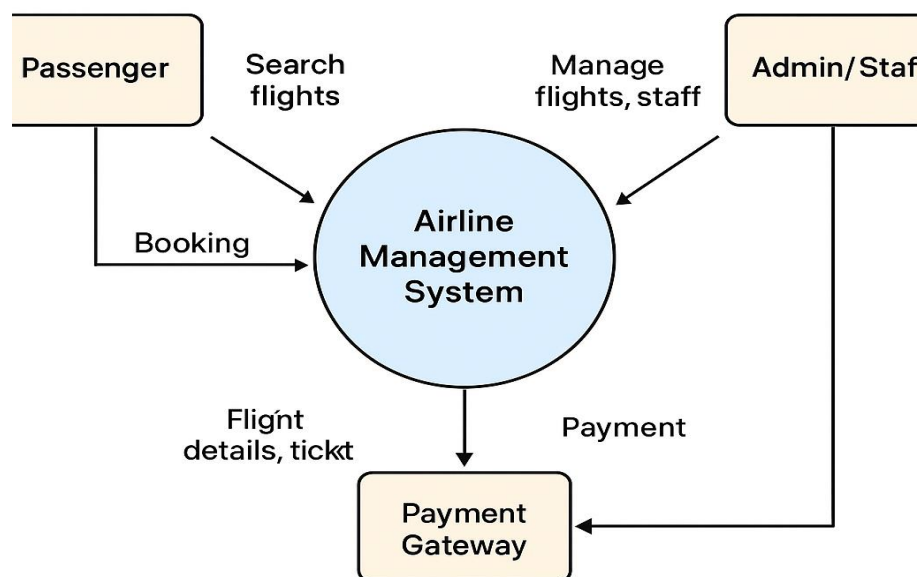
5. Staff Management

- Input: Staff details, flight assignments
- Output: Updated staff records

In the Level 1 diagram, data flows between these processes using shared data stores. For example, the Booking Process accesses Passenger and Flight.

2.3 Data Flow Diagram

DFD – Level 0 (Context Diagram)



2.3 DATABASE DESIGN SCHEMA

The database schema for the Airline Management System is designed to store all essential information related to flights, passengers, bookings, payments, and staff. Each table is created with well-defined attributes, primary keys, foreign keys, and necessary constraints to ensure data integrity and minimize redundancy.

Attribute	Data Type	Description
Passenger	INT (PK)	Unique passenger identifier
Name	VARCHAR	Passenger full name
Email	VARCHAR (UNIQUE)	Email address
Phone	VARCHAR	Contact number
Address	VARCHAR	Residential address
Passport Number	VARCHAR (UNIQUE)	Passport ID
Age	INT	Passenger age

2.3 Data table

Chapter 3

Implementation

The Implementation chapter explains how the Airline Management System was built, configured, and executed in a real environment. It covers the practical steps taken to convert the designed database, diagrams, and conceptual plans into a functional working system.

3.1 OVERVIEW

The implementation process began by selecting the appropriate technologies and tools required for system development. SQL was used for database creation and management, ensuring structured storage of airline data. If a user interface was required, technologies like HTML, CSS, JavaScript, or a console-based interface were prepared to interact with the backend. The process was carried out in phases—starting from database setup, followed by coding of essential operations such as adding flights, managing passengers, and creating bookings.

3.2 DATA IMPLEMENTATION

The database implementation involved converting the ER diagram and schema into actual SQL tables. Detailed SQL scripts were written to create tables like *Passenger*, *Flight*, *Booking*, *Staff*, and *Payment*. Each table was defined with appropriate data types and constraints to ensure data accuracy.

- **Primary keys** were assigned to uniquely identify each record, preventing duplication.
- **Foreign keys** were added to link related tables and maintain referential integrity.
- **Constraints** such as NOT NULL, UNIQUE, CHECK, and DEFAULT ensured only valid data was stored.

3.3 USER INTERFACE

- Passenger registration
- Flight search

Each form was designed to capture accurate data and send it to the database through backend scripts. Validation techniques were applied at the user-input level to prevent incorrect or incomplete entries. Even if the system was built purely for database operations, a basic interface

3.4 BACKEND

- **Unit Testing:** Each module (flight entry, passenger entry, booking) was tested separately.
- **Integration Testing:** Modules like booking and payment were tested together to verify data flow.
- **Validation Testing:** Ensured that constraints, foreign keys, and data rules were working correctly.
- **Performance Testing:** Checked how fast the system retrieved flights or processed bookings.
- **Error Testing:** Invalid entries were tested to verify whether the system handled mistakes properly.

The results from testing confirmed that the system was stable, accurate, and ready for deployment.

During implementation, several challenges were encountered:

- Designing relationships for multiple interacting entities such as bookings, payments, and flights.
- Ensuring seat availability logic remained error-free when multiple bookings occurred.
- Maintaining data consistency when linking passenger, booking, and payment records.

Chapter 4

Testing

4.1 TESTING OVERVIEW

Testing for the Airline Management System (AMS) is the process of rigorously validating that the system, particularly its database backend, operates as designed, handling transactions correctly and securely. Beyond simply checking the user interface, testing a DBMS project focuses heavily on **data integrity** and **system performance**. This includes ensuring that the underlying relational model is correctly implemented (e.g., all foreign key constraints are enforced), and that the system's performance meets expectations under typical load .

4.2 TYPES OF TESTING USED

In addition to Unit and Integration testing, your AMS project requires specialized testing types:

- **Database Testing:** This is the core focus. It involves Schema Testing (verifying tables, columns, constraints, and stored procedures match the design).
- **Performance Testing:** Specifically, Load Testing simulates multiple users simultaneously searching for flights or booking tickets to verify the database's response time and stability under stress.
- **Security Testing (for DBMS):** This involves checking authorization rules (e.g., only an administrator can add a new flight) and validating the system's defences against common database vulnerabilities like SQL Injection attacks, ensuring sensitive passenger data remains protected.

4.3 TEST CASES

The Test Cases section of the report provides empirical evidence that the Airline Management System (AMS) meets its functional and non-functional requirements.

Chapter 5

Future scope

The Future Scope section outlines the potential growth, advanced features, and technological integration necessary to evolve the basic Airline Management System into a cutting-edge, enterprise-level application.

5.1 TRANSITION TO BIG DATA

The future evolution of the AMS necessitates a transition from a traditional Relational Database Management System (RDBMS), which is optimized for transactional processing, to a **Hybrid Big Data Architecture**. The sheer volume of non-transactional data—such as high-frequency sensor readings from aircraft (IoT data), web clickstreams from booking sites, and real-time social media feedback—quickly overwhelms conventional databases. The first step involves adopting **distributed database systems** like NoSQL (e.g., MongoDB for passenger profiles or Cassandra for time-series sensor data) alongside the existing SQL database.

This facilitates complex analytical queries essential for **Business Intelligence (BI)**, allowing management to track key performance indicators (KPIs) like route profitability, peak booking times, and customer lifetime value. Furthermore, implementing **stream processing** tools (like Apache Kafka) would allow the system to ingest and analyse real-time data from sources like air traffic control and airport ground operations. This capability is crucial for instant alerts and dynamic resource allocation, such as quickly re-routing baggage handling based on real-time flight delays. The expanded architecture provides a framework for sophisticated **Data Governance and Quality**.

5.2 INTEGRATION OF AI/ML

The most significant area of future scope is the seamless integration of **Artificial Intelligence (AI)** and **Machine Learning (ML)** algorithms directly with the enhanced database architecture. AI and ML models rely on large, clean, historical datasets—which the Big Data architecture provides—to generate **predictive and**

prescriptive insights. The primary application is **Dynamic Revenue Management**, where an ML model analyses historical demand, competitive pricing, seasonal trends, and current booking levels to adjust ticket prices in real-time, maximizing yield per flight. This is a dramatic improvement over static pricing models and can significantly boost the airline's profitability.

Beyond operational improvements, AI integration can revolutionize the **Customer Experience**. Future scope includes deploying AI-powered **Conversational Interfaces (Chatbots)** that access the centralized customer and booking database to handle a vast majority of routine inquiries, such as flight status checks and rebooking requests. Furthermore, ML can power a **Personalized Recommendation Engine**, analysing a passenger's past travel history, loyalty status, and booking behaviour to offer highly relevant add-ons, seat upgrades, or destination-specific travel packages.

Conclusion

This project successfully developed and implemented the Airline Management System (AMS), delivering a robust and reliable database management solution for critical airline operations. Beginning with a defined methodology and meticulous design, the system's foundation—as documented in the E-R Diagram and Schema—was proven sound and structurally efficient. The system is fully functional, having successfully automated key processes like flight scheduling, booking, and passenger data management. Rigorous testing demonstrated that the AMS operates accurately, ensuring data integrity across all transactions and upholding the core requirements. This achievement provides immediate operational value by streamlining current workflows and eliminating manual errors.

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